

G A R V K H U R A N A

LEADER | BUILDER | INVENTOR



TECHNICAL LEADERSHIP & AI/ML EXPERTISE

- . **AI/ML & Generative AI:** Large Language Models (GPT, Claude, Gemini, Llama, Mistral), Retrieval-Augmented Generation (LangChain, LlamaIndex, Haystack), Multi-Agent Systems (LangGraph, AutoGen, CrewAI), ML Frameworks (TensorFlow, PyTorch, Hugging Face Transformers), Model Evaluation (LLM-as-Judge, Custom Evals, RAGAS, A/B testing), MLOps (Weights & Biases, MLflow)
- . **Engineering & Architecture:** Python, JavaScript, TypeScript, SQL, R; Node.js, FastAPI, Flask, React, Next.js; AWS, Azure, GCP, Docker, Kubernetes, CI/CD; PostgreSQL, MongoDB, Redis, Pinecone, Weaviate, ChromaDB, Qdrant
- . **Specialized Domains:** DeFi & Blockchain (smart contracts, protocol analysis, oracle systems), Risk Modeling (actuarial methods, quantitative modeling, Monte Carlo simulations), Data Science (statistical analysis, time series forecasting)
- . **Leadership & Product:** Team leadership (15+ engineers), cross-functional collaboration (10-15 teams), product management, roadmap planning, stakeholder management, P&L ownership, budget allocation

PROFESSIONAL EXPERIENCE

Lead Architect (Generative AI), Fractal.ai
Enterprise AI company serving Fortune 500 clients

August 2023 – Present
Gurugram, India

- . Architected and deployed **cogentiq**, Fractal's flagship multi-agentic AI platform, serving as technical foundation for **5+ production** enterprise applications
- . Leading **15+ member engineering team** across 2-3 parallel AI/ML projects; managing technical roadmap, architecture decisions, and cross-functional collaboration with **10-15 teams**
- . Built **Underwriting Risk Product** as Lead Tech & Product Owner—an AI-powered vertical solution on cogentiq that automates risk assessment for insurance enterprises using LLMs and multi-agent workflows
- . Developed **Marshall Goldsmith AI** (marshallgoldsmith.ai), an AI coaching bot leveraging LLMs for personalized executive coaching and leadership development
- . Led architecture decisions for production LLM applications including prompt engineering, RAG optimization, vector database selection, custom evaluation frameworks, and model deployment strategies
- . Conducted research on **Named Entity Recognition using LLMs vs. traditional ML baselines**, establishing evaluation frameworks and cost-performance analysis for enterprise AI model selection

Founder & CEO, Techmendous Inc.

2018 – 2023

Finance and technology company providing R&D as a service across algorithmic trading, DeFi, and AI *Virtual Office - Global*

- . Generated **\$500K+ revenue** from proprietary algorithmic trading systems; products recognized by Institute of Actuaries of India and Quantinsti for innovative application of actuarial methods
 - . Achieved **Top Rated status on Upwork** with **\$250K+ earnings**; delivered 20+ five-star projects in algorithmic trading, DeFi protocol development, and financial technology consulting
 - . Built and managed **cross-disciplinary teams of 8-10 members** (Python developers, ML engineers, actuaries, full-stack engineers); responsible for P&L management, hiring strategy, and budget allocation
 - . Created **AWS ML/AI educational content** (9 lecture videos) watched by Fortune 500 executives; covered machine learning fundamentals, deployment strategies, and enterprise AI adoption
 - . **DeFi Research & Development (2021-2023):** Developed Euler Risk Framework for DeFi lending protocol security; contributed to [EulerSwap research paper](#); built UniswapV3 Offline Simulator for liquidity modeling and cost-of-attack analysis; integrated FTSO oracle systems; developed Balancer Protocol simulations and Topgun Arbitrage platform
 - . Mentored **2 actuarial analysts** who progressed to senior positions at major financial institutions and research universities
 - . Bootstrapped entire business from concept to profitability within 18 months through technical execution and strategic client acquisition
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Founder & Lead, Personal Projects

Ongoing

AI and DeFi innovation initiatives

- . **aggrag**: Authored open-source framework for RAG experimentation enabling configuration, comparison, and deployment of multiple retrieval-augmented generation pipelines; **presented at NVIDIA GTC 2024** in San Francisco
- . **insightsin5**: Developing AI-powered business intelligence platform for Shopify store owners, leveraging natural language to SQL queries and LLM-based insight generation

Actuarial Analyst, RSA Insurance Group

2014 – 2017

Legacy 300-year-old insurance business managing \$8.5bn pension liabilities

Delhi/London

- . Built quantitative models for pension liability portfolio management; improved forecasting of rare catastrophic events (e.g., Lehman Brothers bankruptcy impact)
- . Trained **8-member analyst team** on quantitative modeling software; revamped technical training curriculum improving onboarding efficiency by 40%
- . Delivered **\$200M+ value** through systematic risk analysis, diversifying between systematic and diversifiable risk components

Risk Analyst, Essex Lake Group

2014

Profit enhancement consulting firm with \$42M annual revenue, founded by Wharton alumni

Delhi

- . Led team to **Top 6 placement** in corporate case challenge at Indian School of Business, formulating 5-year strategic roadmap for Indian government digitization initiatives

SELECT PROJECTS & OPEN SOURCE

- . **aggrag** (NVIDIA GTC 2024): Open-source framework for RAG experimentation; unified pipeline for configuration, comparison, and production deployment. Tech: Python, LangChain, vector databases
- . **cogentiq**: Multi-agentic AI platform at Fractal.ai for orchestrating AI agents across enterprise workflows. Foundation for underwriting risk, customer service, and business intelligence solutions
- . **Marshall Goldsmith AI**: Executive coaching platform leveraging LLMs for personalized leadership development. Tech: LLMs, conversational AI, RAG, custom evaluation frameworks
- . **Euler Risk Framework**: Comprehensive DeFi security research analyzing collateral, liquidity, oracle risks. Includes Euler Scan monitoring tool and cost-of-attack analysis framework
- . **UniswapV3 Offline Simulator**: Protocol analysis tool for liquidity modeling and cost-of-attack analysis in AMM protocols. Tech: Python, Web3.py, mathematical modeling

ACADEMIC EXPERIENCE

Student Actuary

2013 – 2019

Institute and Faculty of Actuaries & Institute of Actuaries of India

UK & India

- . Cleared CT Series; applied actuarial methods to non-traditional domains including algorithmic trading and DeFi protocol risk assessment

Executive Program in Algorithmic Trading, Quantinsti

2016

6-month CPD-accredited foundation course

India

- . Exemplified application of actuarial methods outside traditional insurance; recognized by profession leadership

Automobile Engineering, B.Tech

2009 – 2013

Manipal University

India

- . Founding member of SolarMobil Manipal to design and fabricate solar-powered race car; 4 vehicles launched since 2011

SPEAKING, RESEARCH & RECOGNITION

- . **NVIDIA GTC 2024** (San Francisco): Poster presentation on aggrag framework for RAG experimentation
- . **AWS ML/AI Educational Content**: Created 9 lecture videos on ML/AI for Fortune 500 executive audience
- . **EulerSwap Research Paper**: Contributed to research on novel AMM protocol design (2023)
- . **AI Evaluation Coursework**: Advanced training by Hamel Husain on LLM evaluation methodologies
- . **Top Rated Freelancer on Upwork**: \$250K+ earnings, 20+ five-star reviews
- . **Institute of Actuaries Recognition**: Acknowledged for innovative application of actuarial methods to algorithmic trading
- . **Mentorship**: Guided 2 analysts who progressed to senior roles at major institutions